

Module 17: Maintenance, Monitoring & Logging

Duration: 3-5 days | **Level:** Advanced

Logging

Laravel uses Monolog. Configure in config/logging.php:

```
LOG_CHANNEL=stack
LOG_LEVEL=debug # production: error or warning
```

Write logs:

```
Log::info('Task created', ['task_id' => $task->id]);
Log::error('Payment failed', ['exception' => $e]);
report($exception); // Log + optionally send to external service
```

Log channels: single, daily, slack, papertrail, syslog.

Error Tracking

Service	Integration
Sentry	sentry/sentry-laravel
Flare	spatie/laravel-ignition (included)
Bugsnag	bugsnag/bugsnag-laravel

```
composer require sentry/sentry-laravel
php artisan sentry:publish
```

Application Monitoring

- **Laravel Pulse** - real-time app metrics
- **Laravel Horizon** - Redis queue dashboard
- **New Relic / Datadog** - APM

```
composer require laravel/pulse
php artisan vendor:publish --provider="Laravel\Pulse\PulseServiceProvider"
```

Health Checks

Built-in: GET /up

Custom health route:

```
Route::get('/health', function () {
    return response()->json([
        'status' => 'ok',
        'database' => DB::connection()->getPdo() ? 'connected' : 'failed',
        'cache' => Cache::has('health_check') || Cache::put('health_check', true, 60),
    ]);
});
```

Maintenance Mode

```
php artisan down --secret="bypass-token"  
# Site shows maintenance page; access via /bypass-token
```

```
php artisan up
```

With message:

```
php artisan down --render="errors::503" --retry=60
```

Database Maintenance

```
# Backup  
php artisan db:backup # with spatie/laravel-backup  
  
# Manual MySQL dump  
mysqldump -u user -p taskforge > backup.sql  
  
# Optimize  
php artisan model:prune # Delete old soft-deleted records
```

Spatie Backup

```
composer require spatie/laravel-backup
```

Schedule daily backups to S3.

Performance Monitoring Checklist

- [] Set up error tracking (Sentry)
 - [] Monitor queue depth and failed jobs
 - [] Disk space alerts on server
 - [] Database slow query log
 - [] Uptime monitoring (UptimeRobot, Pingdom)
 - [] SSL certificate expiry alerts
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Log Rotation

Configure daily channel with days retention:

```
'daily' => [  
  'driver' => 'daily',  
  'path' => storage_path('logs/laravel.log'),  
  'level' => 'debug',  
  'days' => 14,  
],
```

Exercises

1. Add Sentry or configure Slack log channel for errors.
 2. Create custom Artisan command app:health-check for monitoring.
 3. Set up daily database backup schedule.
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Next Module

-> [Module 18: Advanced Topics & Architecture](#)